

# Data Verification Log

NovaTech Revenue Intelligence Dashboard | Pre-transformation data quality checks

**Purpose.** Every source was verified against the supplied data dictionary before any transformation was applied, using natural-language queries against the indexed data. Each question targets a fact that can be proved: a row count, a distinct value count, a date range or a known null count. Eight checks were run across all three knowledge bases. Seven passed outright; one returned a partial answer for a reason worth recording.

## Verification log

| # | Knowledge base      | Question asked                                                                    | Expected answer (data dictionary)                                                                                                      | Q's actual answer                                                                                                             | Match?  | Notes                                                                                                                                                                                                                                                         |
|---|---------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | CRM Deals           | How many records are in the CRM deals data?                                       | 499 rows                                                                                                                               | The dataset contains 499 records                                                                                              | Pass    | Also the load check, confirming the file imported completely.                                                                                                                                                                                                 |
| 2 | CRM Deals           | How many deals were won and how many were lost?                                   | Won 315, Lost 184, total 499                                                                                                           | Same split returned with the full breakdown                                                                                   | Pass    | Confirms deal_stage holds exactly two values with no nulls.                                                                                                                                                                                                   |
| 3 | Marketing Campaigns | How many rows have a missing annual_income?                                       | 24 rows, 1.1%                                                                                                                          | 24 rows with a null annual_income                                                                                             | Pass    | Known gap documented in the dictionary. Left as null rather than imputed, since imputing income would distort segment analysis.                                                                                                                               |
| 4 | Marketing Campaigns | How many leads have campaign_response = 1, and what is the overall response rate? | 609 of 2,240 leads, 27.2%                                                                                                              | Total leads 2,240, responders 609, response rate 27.2%                                                                        | Pass    | Response rate reproduced independently before it was built as a KPI card.                                                                                                                                                                                     |
| 5 | Support Tickets     | How many support tickets have no ticket_resolved_date?                            | 59 tickets, 2.0%                                                                                                                       | 59 tickets with no resolved date                                                                                              | Pass    | Cross-checked against an independent profile of the raw file: 3,000 total minus 2,941 resolved equals 59.                                                                                                                                                     |
| 6 | Support Tickets     | How many tickets are there for each priority level?                               | low 1,500, medium 1,050, high 400, critical 50                                                                                         | Identical distribution returned, total 3,000                                                                                  | Pass    | Confirms the priority field is complete and correctly valued across all 3,000 rows.                                                                                                                                                                           |
| 7 | Reference documents | How many sales reps does NovaTech have, and how are they organised by region?     | 30 reps across East, Central and West, 6 managers reporting to the VP of Revenue                                                       | 30 unique reps, evenly distributed, West 10, East 10, Central 10, with no rep operating in two regions                        | Pass    | Answered from the CRM dataset rather than the reference document. Counts agree with the company background brief.                                                                                                                                             |
| 8 | Reference documents | Who are NovaTech's sales managers, and who do they report to?                     | Six managers, Amanda Foster, Brian Nakamura, David Kim, Michael Chen, Rachel Torres and Sarah Mitchell, reporting to the VP of Revenue | Listed all six with their region and their five reps each. Stated that the dataset does not record who the managers report to | Partial | Names, regions and team sizes verified against the data, two managers and ten reps per region with no overlap. The reporting line exists only in the company background document, which is not indexed in this workspace, and Q said so rather than guessing. |

## What the verification changed

Two findings emerged from this process and both altered how the dashboard was built.

**Opportunity identifiers are not a safe unique key.** A structural probe alongside these checks returned 496 distinct opportunity identifiers against 499 deals, meaning three are reused on different accounts. Any measure built on that field alone would silently under-report. The unique deal key is account and opportunity together.

**The two systems disagree on customer size.** The CRM records four company size tiers while the support system records three. The same customer can therefore carry a different size label depending on which system is asked, which is why segment filtering across the three views is built on an explicitly aligned field rather than assuming the two are interchangeable.

**On the partial result.** Entry 8 is recorded as partial rather than failed because the behaviour was correct. Asked for information that exists only in an unindexed document, the tool answered the part it could evidence from data and stated plainly that the rest was not available, rather than producing a plausible guess. That is the response you want from a system a business team will rely on.

Checks performed against the indexed CRM, marketing and support knowledge bases before any transformation was applied.